Analyst Expectations Provide No Incremental Return Predictability Beyond Valuation*

Felipe S. Iachan and Raul Riva

FGV EPGE

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Abstract

Consensus long-term growth forecasts (LTG) and analyst disagreement do not robustly predict aggregate market returns. Three findings support this conclusion. First, LTG survives conditioning on the spot price-earnings ratio—the control used in recent work on overpricing—but both LTG and disagreement become insignificant once the cyclically adjusted price-earnings ratio (CAPE) is substituted. Second, the LTG–return relationship is unstable across subperiods: coefficients change sign, and the post-2008 regime differs markedly from the pre-crisis period. Third, recursive out-of-sample forecasts across multiple start dates and horizons confirm that neither variable delivers reliable predictive performance. Together, these findings challenge the claim that subjective beliefs retain incremental predictive power beyond valuation ratios: their apparent significance is subsumed by an adequate valuation control, is unstable across subperiods, and is absent out of sample.

Keywords: Subjective beliefs; predictability; out-of-sample; earnings-per-share forecasts; disagreement; CAPE; I/B/E/S;

JEL Classification: G12; G17; G40.

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1 Introduction

Subjective beliefs about long-term earnings and disagreement among forecasters exhibit multiple signs of weakness as predictors of aggregate market returns. We find that the apparent predictive power of consensus long-term growth forecasts (LTG) and analyst disagreement (D) vanishes under closer empirical scrutiny. Three findings support this conclusion. First, in the in-sample horse race that the literature uses to support overpricing mechanisms, replacing the spot PE ratio with the cyclically adjusted price-earnings ratio (CAPE) renders both LTG and D insignificant. Second, even the unconditional LTG–return relationship is unstable across subperiods, with coefficients that change sign and a post-2008 regime that differs markedly from the pre-crisis period. Third, recursive out-of-sample forecasts across multiple start dates and horizons show that neither variable delivers reliable predictive performance.

These empirical results challenge a central piece of evidence used to support theories of extrapolation and overpricing: that subjective beliefs have robust incremental power to forecast returns. In extrapolation models, excessive optimism about earnings growth fuels high prices and disappointing subsequent returns, so a negative relationship between LTG and future returns is often presented as supporting evidence (La Porta, 1996; Bordalo et al., 2019, 2024). Similarly, in overpricing models building on Miller (1977), high disagreement, combined with short-selling constraints, allows optimists to bid prices up unchecked, and negative return predictability by D is used to support this (Yu, 2011). Because LTG, D, and valuations co-move, this literature frequently relies on tests of whether belief-based measures offer incremental predictive power beyond valuation ratios. We jointly reassess these two strands and find that their predictive power is fragile: it is subsumed by an adequate valuation control, unstable in sample, and unsuccessful out of sample.

These results bear directly on recent empirical evidence used to support the theories above. Bordalo et al. (2024) propose a model of belief extrapolation and show, as supporting evidence, that LTG retains significant predictive power for aggregate returns after conditioning on the spot price-earnings ratio. We revisit this evidence using CAPE—a more standard valuation control in the forecasting literature—extending the sample through 2022, and evaluating performance out of sample. The incremental predictability of LTG does not survive any of these checks. Similarly, the robust negative effect of D documented by Yu (2011) can be replicated on his pre-2008 sample but does not hold once the sample is extended or CAPE is included. The answer is measurement: the spot PE ratio is a noisier valuation control than CAPE, and this noise creates room for belief variables to appear significant. Once the valuation control is improved, the apparent incremental content vanishes.

Two important qualifications are in order. Our findings do not falsify theories that establish that LTG is relevant for *price determination*. The strong contemporaneous correlation between LTG and CAPE—which we confirm—is consistent with the view that analyst expectations are reflected in current stock prices. What we show is that this very correlation leaves LTG with no *incremental* information for predicting future returns beyond what the valuation ratio already captures. In the framework of Nagel and Xu (2023), survey-based expectations update sluggishly relative to market prices, so valuation ratios—which aggregate information from many participants—dominate in forecasting regressions. Weak predictive power does not imply that subjective beliefs are irrelevant for prices; it reflects the information limitations of survey expectations relative to market-price-based measures.

Beyond LTG, we examine analyst disagreement as a second belief-based predictor. Where LTG captures the extrapolation channel—optimism drives prices above fundamental value—D captures the overpricing channel of Miller (1977), in which short-sale constraints prevent pessimists from correcting inflated prices. Testing both channels jointly sharpens the conclusion: neither LTG nor D retains predictive power once valuations are properly measured, suggesting that the information in subjective expectations is already reflected in CAPE.

Relationship with literature. Our paper speaks to several strands of the asset pricing literature.

A recent surge of interest in measuring market participants' subjective beliefs has emerged, with these measures used to account for observed asset price dynamics. Incorporating subjective beliefs leads to conclusions that substantially diverge from those of representative-agent rational-expectation models. For instance, empirical subjective beliefs are volatile enough to account for the price-earnings ratio variability seen in the data (De La O and Myers, 2021; Bordalo et al., 2024), and there is no evidence of a counter-cyclical subjective risk premium (Nagel and Xu, 2023). More closely related to us, La Porta (1996) and Bordalo et al. (2019) show how LTG from I/B/E/S can help price the cross-section of stocks, and Bordalo et al. (2024) extends this to the aggregate market. Our work tests whether this aggregate relationship survives appropriate valuation controls and out-of-sample evaluation.

The literature on heterogeneous beliefs and overpricing, building on Miller (1977) and Jarrow (1980), predicts that disagreement should negatively forecast returns in the presence of short-selling constraints. Yu (2011) provides empirical support using analyst disagreement, but the analysis is in-sample and stops before 2008. We extend this evidence to a longer sample and show that the predictive power of disagreement is fragile once CAPE is included and the sample

is extended.[1]

Hong and Sraer (2013) provide a theoretical taxonomy distinguishing "loud" equity bubbles from "quiet" credit bubbles: disagreement affects equity prices strongly (linear payoffs, large resale option) but has a limited impact on debt prices (bounded upside suppresses the resale option). Their 2016 paper (Hong and Sraer, 2016) uses a measure of earnings disagreement to confirm "speculative beta" predictions cross-sectionally. Neither paper studies the time-series behavior of I/B/E/S -based disagreement or its evolution over time. The behavioral shift we document in equity-analyst disagreement after 2008 occurs on the equity side of their taxonomy, where disagreement should remain relevant, and is not predicted by the quiet-bubbles mechanism.[2]

Nagel and Xu (2023) document that investors' subjective expected returns update sluggishly relative to rational-expectations benchmarks. If analyst expectations of fundamentals are subject to similar informational rigidities and incorporate information slowly, they will be correlated with true expected returns but will lag market-price-based measures that aggregate information more efficiently. Our finding that `LTG` loses predictive power once `CAPE` is included is consistent with this evidence.

Huang et al. (2021) take a different approach to measuring disagreement, aggregating analyst, survey, macro, and option-implied sources into a single index via partial least squares. Their composite measure predicts returns in sample and out of sample, with predictive power concentrated in high-sentiment periods. Our analysis is narrower in scope: we focus exclusively on I/B/E/S -based analyst forecasts to isolate the specific belief channel emphasized by Yu (2011), and we test whether this channel retains incremental power once a cyclically adjusted valuation control is included. That our analyst-belief measures lose predictive content against `CAPE` does not speak to the broader aggregation result in Huang et al. (2021), whose composite index may derive its forecasting power from the combination of diverse information sources rather than from any single component. As we show in Section 3, our analyst-belief measures are also empirically distinct from broad sentiment indices, confirming that `LTG` and `D` are not simply repackaging market-wide sentiment.

[1] See also related theoretical work by Atmaz and Basak (2018) and Iachan et al. (2021), and the survey by Simsek (2021).

[2] Cheng et al. (2014) provide complementary evidence that the approaching housing-market correction was largely unanticipated by securitization-industry professionals, whose personal portfolios show no systematic reduction in housing exposure. The message that pre-crisis professional expectations missed the correction is consistent with the post-2008 shift we document, though their setting—personal portfolio choices of securitization staff—is a few steps removed from aggregate analyst forecasts of firm-level earnings growth.

Goyal et al. (2024) survey 29 predictive variables from 26 studies published after Welch and Goyal (2008), evaluating each in sample and out of sample through 2021. Analyst disagreement, as in Yu (2011), is one of the variables they consider, but belief-based measures receive no special emphasis in their analysis. Our paper is complementary in scope: we concentrate on the first and second moments of analyst long-term growth forecasts—LTG and disagreement—and subject them to a battery of tests that includes valuation controls, subsample stability, and out-of-sample evaluation. Our finding that neither variable survives these tests is consistent with the broad weak-predictability pattern that Welch and Goyal (2008) and Goyal et al. (2024) document for return predictors more generally.

The remainder of the paper is organized as follows. Section 2 discusses our data sources and defines the main variables. Section 3 details the construction of both the consensus and disagreement measures. Section 4 presents our in-sample results; out-of-sample analyses are discussed in Section 5. Section 6 concludes.

2 Data

Our data come from three different sources. First, we use CRSP data to collect end-of-month closing prices and the number of outstanding shares. Our notion of market return here is also derived from the value-weighted index available through CRSP.[3] This data delivers the main target we try to predict and the respective market capitalization weights we use throughout the paper. As usual, we consider stocks that traded at the NYSE, AMEX, and NASDAQ. We also focus on common stock, using share codes 10 and 11 from CRSP. Our notion of asset returns is the arithmetic return, regardless of the horizon.

We also use data from I/B/E/S to measure analyst forecasts.[4] Since March 1976, I/B/E/S has collected professional analysts' forecasts for many different performance measures for firms, such as earnings per share, return on investment, return on equity, etc. The prime forecast variable in I/B/E/S is, however, earnings per share, and that's the one we concentrate on. This is the variable with the best coverage on I/B/E/S , both in terms of time series and cross-sectional availability. Also, it is less prone to endogenous choices from the firm's management side, which we do not aim to model, such as dividends per share. Analysts are asked to forecast firm earnings at different horizons on a rolling basis, ranging from the subsequent fiscal quarter

[3]The correlation with S&P500 returns is over 99% at the monthly horizon. We experimented using S&P500 returns, and the results were nearly identical. For the sake of space, we don't report those, although they are available upon request.

[4]For a recent discussion of I/B/E/S-related measures, see Adam and Nagel (2023).

up to a long-term earnings growth forecast. These forecasts are collected (and updated) on a monthly frequency.[5]

Using this long-term earnings growth forecast over other I/B/E/S measures has at least three advantages. First, analysts report a growth *rate* for the long-term forecast while they forecast *dollar amounts* in other cases. When using rates, we do not need to worry about, for instance, a numeraire. In contrast, due to inflation, dollar amounts from different periods need to be normalized. These forecasts based on rates are, therefore, more comparable over time. Second, this long-term forecast, in theory, should be less influenced by short-term fluctuations in the business cycle and reflect a cleaner assessment of growth opportunities for the firms. Third, we use data that is comparable to previous literature.

We also acknowledge an important unfavorable aspect of long-term forecasts. Analysts are required to forecast an annual growth rate over a period that is, unfortunately, loosely defined. The precise definition of the rate they should forecast is *"... the expected annual increase in operating earnings over the company's next full business cycle. These forecasts refer to a period of between three to five years."*. We understand that different analysts might have slightly different horizons in mind when making those forecasts. To remedy this issue, aside from noting that analysts report annual rates anyway, we examine forecasting performance at shorter and longer horizons and show that our results are consistent across these horizons. We also note that long-term forecasts are available to us starting in December 1981, and that's the starting point of our empirical analyses in the next sections. The last month included is December 2022. That amounts to 493 monthly observations.

The last source of data is Robert Shiller's website, from which we collect `CAPE` for the S&P500 index.[6] `CAPE` is defined as the ratio of the current real price of the S&P 500 index to the average of the past ten years of real earnings per share. By averaging earnings over a full decade, this measure smooths out transient fluctuations in earnings caused by the business cycle, producing a more stable valuation benchmark than the standard (spot) price-earnings ratio. This matters for our analysis: the spot PE ratio inherits short-run earnings volatility that can attenuate its predictive power, potentially leaving room for other variables—such as `LTG`—to appear significant in forecasting regressions. `CAPE`, by contrast, provides a cleaner valuation signal and is a standard variable in the return-prediction literature.[7] Throughout the paper, we use `CAPE` to

[5] We note that we are silent about non-US firms. We believe, however, that investigating international evidence is an important step in this agenda.

[6] Available at `http://www.econ.yale.edu/~shiller/data.htm`.

[7] See, for example, Campbell and Shiller (1988), La Porta (1996), Welch and Goyal (2008), and Yu (2011). Jivraj and Shiller (2017) provide a detailed discussion of CAPE's properties.

refer to this cyclically adjusted measure and reserve "spot PE" for the standard (unadjusted) price-earnings ratio.

3 Constructing the Variables

3.1 From CRSP

We use the value-weighted US CRSP index as our notion of index prices to construct returns. We use end-of-month prices, already accounting for dividends. We denote by $R_{t+h|t}$ the h-month arithmetic index return, whose price at time t is denoted by S_t:

$$R_{t+h|t} = \frac{S_{t+h} - S_t}{S_t}. \tag{1}$$

We work with this variable at a monthly frequency, although we study predictability for different values of $h > 1$. That implies we work with overlapping returns. We know this choice induces a moving-average component in linear regression error terms and try to remedy that using HAC-adjusted inference with different lag lengths below.

3.2 From I/B/E/S

All variables used in the analyses evolve at the monthly frequency. Our construction follows standard procedures along the lines of Yu (2011), Huang et al. (2021), and Bordalo et al. (2024), for example. In any given month, we consider firms from I/B/E/S for which we have at least three analysts reporting forecasts. Increasing the threshold for the number of analysts implies that averaging over different forecasts will be less noisy, but it does introduce larger survivorship bias. This bias occurs because a higher number of analysts typically corresponds with larger firms. Given that our analyses weight firms by market value, we deliberately include a broader array of firms.

Then, for each firm i at month t, we track the average over forecasts, which we denote by $LTG_{i,t}$. We also track the standard deviation of these forecasts across analysts, which is a measure of the disagreement, denoted by $D_{i,t}$. We winsorize both measures for each t at the 1% and 99% quantiles to offset the influence of possible outliers. Moreover, we use CRSP to collect their closing price on the last trading day of the month t, denoted by $P_{i,t}$ and the number of

outstanding shares $Q_{i,t}$. We then define the weights $w_{i,t}$ as

$$w_{i,t} \equiv \frac{P_{i,t} \cdot Q_{i,t}}{\sum_{j=1}^{N_t} P_{j,t} \cdot Q_{j,t}},$$

where N_t is the number of firms that survived the first stage. Finally, we define the long-term earnings growth consensus LTG_t and the disagreement about this measure D_t as

$$LTG_t \equiv \sum_{i=1}^{N_t} w_{i,t} \cdot LTG_{i,t} \tag{2}$$

$$D_t \equiv \sum_{i=1}^{N_t} w_{i,t} \cdot D_{i,t} \tag{3}$$

3.3 Summary Statistics and Time-Series Evolution

Now we analyze the basic features of our working variables. Throughout the paper, we use `LTG`, `D`, and `CAPE` to designate information related to our measure of consensus earnings growth, the associated disagreement, and the cyclically adjusted price-earnings ratio, respectively. Similarly, we reserve the notation R_h to denote information related to the h-month market index return.

Table 1 reports summary statistics for the full sample. The first four rows display results for market returns, while the bottom three are dedicated to `CAPE` and the variables we construct from subjective beliefs. It is useful to note that these variables have different scales. This will motivate some scaling in our empirical analysis to make results comparable across horizons and regression specifications.

[Table 1 here]

Figure 1 shows the evolution of `LTG`, `D`, and `CAPE` over time. We demean each variable and divide the resulting series by its standard deviation to have the three variables on the same scale.[8] The shaded bars denote NBER recessions. There are several interesting aspects of their joint evolution.

[Figure 1 here]

[8] In Appendix A, Figure A.3 plots all variables in individual panels at their original scales for convenience.

We discuss the evolution of CAPE and then focus on the subjective belief variables. There has been an almost monotonic increase since the early 1990's until its peak just before the dot-com bubble burst in early 2000. It then follows a downward trend, stabilizing its value around 2005, just before a marked dive during the global financial crisis in the late 2000s. The subsequent movement is a gradual recovery that does not seem too disturbed by the Covid 19 pandemic. The very last part of the sample displays a decreasing profile for CAPE, likely driven by lower equity prices in response to the monetary cycle in the US.

The evolution of LTG is similar, although with some important differences. It also peaked just before the dot-com bubble, after a decade of steady increase, followed by a rapid decline, stabilizing around 2005. We also see a sharp decrease in 2008 and some recovery thereafter. On the other hand, during the first nine years of the sample, the consensus growth forecast decreased over time, while CAPE was increasing. During the very last part of the sample, LTG follows a jagged path with no clear trend.

Interestingly, D seems to follow a similar qualitative path until 2008, when we spot some stronger disconnection between LTG and D. Since then, there has been no clear trend for D. It starts moving in tandem with LTG once again towards the very last part of the sample. It is out of the scope of this paper to investigate why this disconnection happens, but this behavioral shift is notable and consistent with other evidence of structural changes in equity-analyst expectations around the financial crisis.

[Figure 2 here]

Figure 2 displays the 120-month rolling correlation between these measures. It makes our points about co-movement more formally. We index the rolling correlations by the final month included. Hence, the curves start in December 1991. The first panel studies the correlation between D and CAPE. It is initially negative and steadily increases, becoming strongly positive just before the dot-com burst. After 2008, the jagged path for D from Figure 1 is translated to a lower correlation with CAPE. The unconditional correlation between them is 0.36 over the full sample.

The second panel studies the correlation between LTG and D over time. We see a similar profile to the first panel, although the correlation starts positive. In both cases, it peaked just before the dot-com bubble burst. After 2008, the correlation changes sign for some time, which is another way of visualizing the disconnection between LTG and D after 2008. The unconditional correlation between them over the full sample is 0.30.

The last panel shows an important stylized fact. Aside from some initial rolling windows

when correlation was negative, it is always positive for the (LTG, CAPE) pair, and strongly so. This can be seen in Figure 1 by how movements in the red line co-move with the green one. The correlation starts negative because, for the first ten years of data, the increase in CAPE happened at the same time when the consensus growth forecast was becoming more pessimistic. The correlation is reduced, although still positive, during the last part of the sample due to the jagged path followed by LTG, which was accompanied by a relatively smooth evolution of CAPE. The unconditional correlation between these two variables is 0.68.

3.4 Relationship with Sentiment Indices

A natural question is whether the belief measures we construct from I/B/E/S simply proxy for broad market sentiment. To address this, we compute correlations between LTG, D, CAPE, the Baker and Wurgler (2006) orthogonalized sentiment index, and the PLS investor sentiment index from Huang et al. (2021). Table A.1 reports the full matrix over the overlapping sample (December 1981 – December 2020, $N = 469$ months).

LTG has a moderate Pearson correlation with BW sentiment ($r = 0.41$) and a low correlation with the HLW index ($r = 0.36$). D is *negatively* correlated with BW sentiment ($r = -0.22$), consistent with the predictions in Atmaz and Basak (2018). CAPE is essentially uncorrelated with either sentiment index ($|r| < 0.18$). The two sentiment indices are themselves nearly uncorrelated with each other ($r = 0.10$), underscoring the conceptual distinction between market-based sentiment proxies and analyst-forecast-based belief measures. These patterns confirm that our analyst-belief measures are empirically distinct from broad sentiment: while the moderate LTG–BW overlap is expected—optimism in analyst forecasts naturally co-moves with market-wide sentiment—it leaves more than 80% of the variation unexplained.

4 In-Sample Analysis

Our in-sample analysis proceeds in two steps. First, we replicate the main specifications from prior work—in particular Bordalo et al. (2024) and Yu (2011)—using the same variables and sample periods. This confirms that we can reproduce their published results. Second, we substitute the cyclically adjusted price-earnings ratio (CAPE) for the spot PE ratio used in those papers and examine how the results change. As we show, this single substitution is enough to render the belief variables insignificant.

Previous work motivates negative coefficients on both disagreement and long-term growth

forecasts in return-predictive regressions. In the disagreement literature, Yu (2011) tests the market-index implications of Miller (1977) and finds evidence consistent with disagreement-fueled overpricing in data through 2007.[9] In the extrapolation literature, Bordalo et al. (2024) argue that LTG predicts aggregate returns in data through 2015. Because LTG, D, and CAPE co-move over time, our in-sample analysis studies the two belief variables jointly and across the three natural samples in this literature: the full sample through 2022, the Bordalo et al. (2024) sample through 2015, and the Yu (2011) sample through 2007.

We study predictive regressions of the following form:

$$R_{t+h|t} = \alpha + \beta_{LTG} \cdot LTG_t + \beta_D \cdot D_t + \beta_{CAPE} \cdot CAPE_t + \epsilon_{t+h} \tag{4}$$

where $R_{t+h|t}$ is the market return over h months. We focus on $h \in \{12, 36, 60\}$ for the in-sample results. When making inferences about estimated coefficients, we use an HAC-adjusted standard error estimator along the lines of Newey and West (1987), with h lags. Since these variables have different scales, as shown in Table 1, we demean each and divide them by their respective standard deviations. The coefficients then measure the effect on the dependent variable due to a one-standard-deviation increase in each variable on the right-hand side, expressed in standard deviations of the left-hand-side variable.

4.1 Main Results

Table 2 summarizes our in-sample results. Each of the three panels concentrates on a different sub-sample. Each panel reports estimates for 18 regressions since there are six specifications for each horizon h. Regressions with the same dependent variable are grouped. In all cases, we report t-statistics in the parentheses. Stars measure statistical significance at 10%, 5%, and 1%, respectively.

In each of these groups, the first specification studies the effect of including only LTG to (4). The second specification includes only D. The third specification uses both the consensus growth forecast and the disagreement measure to predict returns. Its main goal is trying to disentangle the effects of these two variables. The fourth, fifth, and sixth specifications follow the same pattern, but all control for the price-earnings ratio as well. They are designed to assess the effects of the variables constructed from subjective beliefs on market returns *above and beyond* what can already be accounted for by a simple measure of stock market overvaluation.[10] As

[9] See also Jarrow (1980) and, for a more recent treatment, Atmaz and Basak (2018).

[10] See Campbell and Shiller (1988) for this interpretation regarding the price-earnings ratio.

evidence for overpricing mechanisms, both literatures would point to coefficients on LTG and D that are negative and significant.

[Table 2 here]

4.1.1 Full Sample (1981-2022)

We concentrate on Panel A. At the 12-month horizon, the coefficients on LTG and D have the sign one would expect from reading previous literature. Increases in these variables are associated with lower future returns. However, the effect through LTG is insignificant after controlling for the price-earnings ratio. The effect of D remains significant, although only at the 10% level, with a t-statistic of -1.75.

At the 36-month horizon, the individual coefficients have the expected sign, but none are significant. That happens both with and without the control provided by the price-earnings ratio. Only the coefficients on the price-earnings ratio are significant, with the correct expected negative sign. The inclusion of neither LTG nor D boosts the in-sample R^2 in a meaningful way.

The 60-month horizon follows the same pattern. Although LTG remains negative and significant after the inclusion of D, controlling for CAPE reverses its sign, though it is not statistically different from zero. We also notice that the inclusion of CAPE makes the R^2 increase more than twofold, indicating a much stronger in-sample predictive power of CAPE relative to the two survey-extracted belief moments.

4.1.2 The Sample Up to December 2015

We now shift our focus to Panel B. This panel mimics the sample from Bordalo et al. (2024), stopping in December 2015. We start emphasizing specifications 1, 7, and 13. These are the basic specifications reported in Bordalo et al. (2024), and we confirm that we can qualitatively replicate their results and almost match their estimates. Our estimates for β_{LTG} range from -0.32 to -0.47, while theirs go from -0.23 to -0.43. We also report similar R^2 measures, validating our construction.

In this sample, D has a significant coefficient only at the 12-month horizon, although it still has the expected sign over the other two horizons. When we include both LTG and D, without controlling for CAPE, the estimates for β_{LTG} generally decrease in absolute value, but such movement is not strong enough to hinder them insignificant.

The inclusion of CAPE in these regressions changes the results, however. When controlling for CAPE, no coefficient was significant at the 12-month horizon. When considering the other two horizons, only CAPE itself is significant. The coefficient on D remains negative across specifications, but we can never reject that it is different than zero. At the 36-month horizon, the estimates for β_{LTG} are essentially zero, while they become positive at the 60-month horizon. Although we confirm the initial finding from Bordalo et al. (2024), we show that controlling for the cyclically adjusted price-earnings ratio makes estimates insignificant or even flips their sign.[11]

4.1.3 The Pre-Crisis Sample (1981-2007)

Finally, we analyze Panel C. The sample that ends in 2007 is special in two ways. First, it is the same time frame used by Yu (2011). This will exclude the Global Financial Crisis and the coronavirus pandemic from the analysis. Second, this is the period of highest co-movement between LTG and D, as seen in Figure 2.

In his empirical work, Yu (2011) studied specifications analogous to columns 6, 12, and 18. He included both the consensus growth forecast and spot PE in his analysis. We start stressing that we can replicate his main findings. The coefficient on D is negative and statistically significant even if we control for the two other variables. It is also significant when only D is included in the regressions, as demonstrated by columns 2, 8, and 14. The magnitude of these estimates is also similar across different horizons.[12]

Nonetheless, we have a more surprising result for $\widehat{\beta}_{LTG}$ in this sample. Even though it remains negative and significant when included alone in these regressions, its sign becomes positive across the three horizons when all variables are included. In fact, for the 36-month and the 60-month horizons, it is positive and statistically different from zero, contradicting the findings from a later sample from Bordalo et al. (2024). When LTG is included alongside D, its coefficient is still negative but never statistically different than zero.

[11] In their analysis, Bordalo et al. (2024) do not include D in their regressions, given their focus on a representative-agent framework. They do include a spot PE variable, which is not cyclically adjusted (hence more volatile and subject to sharp changes) and is less standard in the forecasting literature. We compare these two measures and some minor differences in implementation in Appendix C.

[12] Yu (2011) did not study the 60-month horizon originally. Therefore, we can provide a new finding and extend his result to a longer forecasting horizon.

4.1.4 Summary of Results

We can summarize our in-sample results in the following way:

1. The strength of the association between LTG and market index returns highly depends on whether the price-earnings ratio is included or not in these regressions. Even when it has the predicted negative sign, estimates are not statistically different than zero if CAPE is included. Moreover, when both CAPE and D are included, estimates for β_{LTG} are generally positive. They are, sometimes, statistically different than zero - but this is the opposite sign from the belief overreaction mechanism advocated in extant literature.

2. The robust negative effects on market returns documented by Yu (2011) can be replicated, but only on his sample. That effect seems to be much weaker when we consider an extended sample.

3. Controlling for D does not typically hinder the estimates on β_{LTG} insignificant, except for the earlier sample that stops in December 2007. However, it does decrease the absolute value of estimates across samples and horizons.

4.2 Is CAPE a "bad control"?

After inspecting our in-sample results, one could ask: should we include CAPE in these regressions after all? One reasonable concern is that CAPE might only mediate the underlying effect on returns of either the consensus growth forecast or the associated disagreement. More concretely, one could be worried that CAPE is solely a noisy proxy for either of the two variables constructed from subjective beliefs and brings no additional information. In that case, adding this control to regressions could crowd out the effect of either LTG or D and make us incapable of rejecting the null of zero effect directly through subjective beliefs, even when it is false. Borrowing language from the causal inference literature, should we be concerned with CAPE being a "bad control"?[13] This is an important issue both for our analysis and for the literature in general since regressions like the ones presented in Table 2 are usually taken as evidence in favor (or against) different theoretical models.

We believe that both Figure 1 and Figure 2 provide some evidence that the answer is "no". An observer with only the first decade of our data would likely conclude that the consensus

[13] See Angrist and Pischke (2008) and Cinelli et al. (2022) for treatments of bad controls in the standard context of causal inference with cross-sectional data.

growth forecast and CAPE move in different directions. She would draw the same conclusion regarding the disagreement measure. Another observer would conclude, however, that these measures are strongly positively correlated if she were given data from 1990 until the financial crisis. A third observer, inspecting the 2010s, would lean toward saying that they are positively correlated, although not as strongly as during the preceding twenty years. In summary, there is evidence of long-lasting lower frequency movements that could not take place if CAPE were a simple proxy of any of the other measures.

Table 3 quantifies the degree of linear association between CAPE and the measures constructed from subjective beliefs using our full sample. We consider the following contemporaneous specifications, both in levels and in 12-month differences:

$$CAPE_t = \alpha + \gamma \cdot LTG_t + \delta \cdot D_t + u_t \tag{5}$$

$$\Delta CAPE_t = \tilde{\alpha} + \tilde{\gamma} \cdot \Delta LTG_t + \tilde{\delta} \cdot \Delta D_t + \tilde{u}_t \tag{6}$$

where, for any variable x_t, we define $\Delta x_t \equiv x_t - x_{t-12}$. The first specification assesses the degree of linear relationship between CAPE and the variables constructed from subjective beliefs. The second specification, concentrating on 12-month changes, checks whether swings in LTG and D are linearly transmitted to CAPE. This last specification is also useful because these time series are fairly persistent at the monthly horizon, and one could be concerned about spurious findings arising from persistence.[14] Before running each regression, we also normalize all variables. The associated t-statistics are reported inside the parentheses. The first column reports the regression of $CAPE_t$ on $CAPE_{t-12}$ as another way to document the persistence in $CAPE_t$ and as a reference for the R^2 levels. The point estimate of 0.85 is highly significant. The R^2 is above 0.78, even though we are conditioning on the price-earnings ratio lagged by a full year.

[Table 3 here]

The second regression includes only LTG. Although the estimate is positive and significant at usual levels, the R^2 is around 45%, even though both CAPE and LTG are aligned in terms of time indexes, showing that the contemporaneous association is not that strong. The third regression uses only D, delivering a low R^2 and an estimate that is not statistically significant at the 5% level anymore. The fourth column uses both LTG and D and shows that estimates barely change for LTG compared to the second column. The coefficient on D is not significant anymore, and the

[14] Figure A.1 in our Appendix A reports the autocorrelation functions of these variables at the monthly frequency, up to 36 months ahead. By far, CAPE is the most persistent, almost integrated. Figure A.2, also in Appendix A, shows the time-series evolution of these 12-month differences, all scaled before plotting to be shown on the same axis.

R^2 of this regression is essentially the same as the one from column (2). The specifications that use the 12-month differences tell the same story. While these results do not rule out that CAPE is a noisy proxy of the other two variables, they do imply that it is *at most* a very noisy one.

Still under the hypothesis that CAPE is a noisy proxy of either LTG or D, we would expect that, in terms of forecasting power, conditioning only on CAPE would generate worse forecasts than using the other variables. After all, predicting with a cleaner signal from either LTG or D should be better than predicting with that signal plus noise. Our next findings provide evidence that this is not the case, which goes against the idea that CAPE is just a proxy for other variables.

5 Out-of-Sample Analysis

Our previous analyses replicated some of the findings from previous literature and showed that they are arguably dependent on the sample used and the controls included. Nonetheless, all these results happen in sample. We now develop an out-of-sample forecasting exercise to more closely examine whether these different variables have forecasting power. We see this exercise as a complement to our previous analysis. Importantly, this type of exercise will make clear *when* each variable (or combination of variables) performs the best. We also inspect the coefficient estimates from (4) over time.

5.1 Forecasting Methodology

We fixed the sample starting date to December 1981, and we consider a recursive forecasting scheme. This is traditional in the macro-finance forecasting literature.[15] It implies that, for each point in time, we estimate the specification in (4) and predict h periods ahead. We keep track of that forecast and move forward by one month. Then, we estimate the specification (4) and predict once again h periods ahead. We continue this until we exhaust our data.

These forecasts, done with data up to time $t-h$, generate forecast errors that are known at time t. We then use the R^2_{OOS} measure from Campbell and Thompson (2008) to assess the

[15] Huang et al. (2021) and Goyal et al. (2024) also use recursive forecasting designs in a context similar to ours, as this is very common in the forecasting literature. For example, Gu et al. (2020) uses a recursive forecasting scheme to predict equity returns, while Bianchi et al. (2020) and Freire and Riva (2023) use this recursive estimation to forecast sovereign bond risk premia. For a general review, see Clark and McCracken (2013).

quality of these forecasts:

$$R^2_{OOS}(t_0, h) = 1 - \frac{\sum_{t=t_0}^{T} \left(R_{t|t-h} - \widehat{R}_{t|t-h} \right)^2}{\sum_{t=t_0}^{T} \left(R_{t|t-h} - \overline{R}_{t|t-h} \right)^2} \tag{7}$$

where t_0 indexes the first out-of-sample date, $\widehat{R}_{t|t-h}$ is a forecast of the market index return made at $t-h$ and $\overline{R}_{t|t-h}$ is a benchmark computed with information available up to $t-h$. In this setting, a natural benchmark is a model that only includes a constant. Hence, its forecasts equal the historical average of market returns for each considered horizon. This measure is also common in the forecasting literature, comparing the forecasting mean-squared error from both models. We highlight that we index the out-of-sample R^2 by t_0 because it will change depending on where one starts the out-of-sample period, which we discuss below.

When this measure is above zero, it implies that the analyzed model can beat the benchmark. On the other hand, it is negative when the model fails against the proposed baseline forecasting method. We also highlight that the out-of-sample R^2 is bounded above by 1 but has no lower bound. It also does not have a measurement unit since it is an index that can assess the quality of different forecasting methods based on a ratio of loss functions.

Aside from the benchmark, we consider seven models in total. Three models include only each of the individual variables `LTG`, `D`, and `CAPE`; then, three models feature each possible combination of two variables; last, a model includes all three. They are all compared against the historical mean of returns.

Our approach is related to, albeit different from, Huang et al. (2021) and Goyal et al. (2024). The former considers out-of-sample predictability of market index returns based on different measures of disagreement, not only the subjective beliefs measure `D`, which we also use. However, they do not examine the predictability of first-moment measures, such as the consensus growth forecast. The latter analyzes how `D` and several other measures predict market returns. However, they do not have variables such as `LTG` and never study regressions with two or more regressors. In this sense, we find additional support for some of their results and see our approach as complementary to theirs.

5.2 A Snapshot of Performance

In this section, we report a snapshot of each model's performance and leave the variation of relative performances over time for the next sections. As is common in forecasting exercises,

choosing the out-of-sample period is not trivial. Given a fixed amount of data, a long out-of-sample period implies a richer testing site for each model, with the caveat that each of these specifications is estimated more imprecisely at the earlier part of the sample since there is less data to run regressions. Estimation uncertainty might then drive results. In contrast, a very short out-of-sample period allows for comparing models that have been more precisely estimated but can only be tested on a handful of data points.

We believe including the dot-com burst in any *in-sample* period is important. Otherwise, endowed only with the data from Figure 1, any model would take `CAPE` and `LTG` almost as ever-growing variables, at least at lower frequencies. Ideally, we want to include at least one large stock market crash in any in-sample period. After the stock market crash in 2000, a recession ensued. Following the official timing from the NBER, the American economy left the recessionary period in November 2001. Since the longest forecasting horizon we study is five years, we would need to start the out-of-sample period at least in late 2006 to ensure all forecasts include these events in their estimation (or "training", borrowing language from the Machine Learning literature) sample.

However, an analyst in late 2001 would not know how firms navigated the recession since balance-sheet data usually takes a few months to be released, and there are different fiscal year-end dates. Considering all this and trying to maximize our out-of-sample period length, we choose July 2007 as the first out-of-sample date. In that way, even for the longest horizon, forecasts are produced with the knowledge of the crash and preliminary information on how firms did during the recession, even allowing for delays in balance sheet data releases for many firms. This sample start also implies that we consider the performance of different models during the global financial crisis, the recovery period, and the first years after the stock market crash in 2020, a challenging range of different economic conditions.

July 2007 is a reasonable benchmark start date, but our conclusion does not depend on that particular choice. Table 4 reports $R^2_{OOS}(t_0, h)$ for four representative start dates, with the last panel corresponding exactly to the July 2007 benchmark. Appendix Figure B.1 in Appendix B extends the same exercise continuously over t_0.

[Table 4 here]

The July 2007 panel already delivers the main message. `CAPE` alone beats the historical-mean benchmark at all four horizons and delivers the strongest R^2_{OOS} in each column. The gains are modest at 6 and 12 months and become more pronounced at longer horizons, especially at $h = 60$. This is the relevant benchmark for the rest of the discussion: once a valuation ratio is

available, the question is not whether belief-based variables can occasionally generate positive R^2_{OOS} values, but whether they can do better than CAPE. In our data, they do not.

The evidence for LTG is weaker and more local. In the July 2007 panel, LTG alone produces small positive values at 6 and 12 months but negative values at 36 and 60 months. The bivariate model with CAPE and LTG remains positive at 6, 12, and 60 months, yet it never improves on CAPE alone. Across the other panels, LTG occasionally posts small positive values at short horizons, but these gains are fragile, disappear at longer horizons, and never come with Clark–West significance stars. In that sense, LTG may sometimes look mildly useful at short horizons, but it does not deliver a robust case for incremental predictability beyond CAPE.

This ranking is also suggestive of the role of CAPE as a control. If CAPE were merely LTG plus additional noise, one would expect the cleaner signal, LTG itself, to forecast at least as well as CAPE. Instead, the opposite ranking appears in Table 4: CAPE is the stronger predictor, and adding LTG to it does not help. We do not want to lean too heavily on this point, since the two variables need not be linked by such a simple measurement-error structure. Still, the evidence goes in the opposite direction of the claim that CAPE is a bad control for LTG.

The evidence for D is weaker still. In the July 2007 panel, D alone is negative at every horizon, and adding D to either CAPE or LTG does not improve the corresponding simpler specification. The same pattern persists across start dates: disagreement sometimes produces tiny positive short-horizon values, but never a convincing or stable improvement over the benchmark. The variable seems to add estimation noise rather than forecasting content in aggregate-return regressions.

Taken together, Table 4 shows that July 2007 is not essential for our conclusion. When Clark–West significance does emerge, it belongs only to CAPE-alone or CAPE-containing specifications. Appendix Figure B.1 shows the same ranking over the full range of starting dates: CAPE is the only model that repeatedly displays economically meaningful positive regions across horizons, LTG has only narrow and unstable short-horizon gains, and D never becomes a robust competitor.

5.3 Evolution of Forecast Errors

Figure 3 complements the R^2_{OOS} evidence by showing when the forecast failures occur over time, not just whether the average out-of-sample fit is positive or negative.

[Figure 3 here]

Starting in 2000 and ending in 2022, Figure 3 displays the evolution of absolute forecast errors. Each column concentrates on a different forecasting horizon. Each row is dedicated to a different forecasting model. There are 28 panels in total. For any model M, for a fixed horizon h, the forecasting error at time t is defined as

$$e_{t,h}^{(M)} \equiv \widehat{R}_{t|t-h}^{(M)} - R_{t|t-h} \tag{8}$$

where $\widehat{R}_{t|t-h}^{(M)}$ is the forecast for market returns from model M, done with information up to time $t-h$. This random variable is only realized at time t. The different panels that compose Figure 3 display the evolution of $\left|e_{t,h}^{(M)}\right|$ as a function of t, plotted as a solid black line. The red dashed line represents the absolute forecast error for the benchmark, i.e., the constant-only model. The red line is the same across different rows since each column has a fixed forecasting horizon. The unit of the vertical axis is the same as market returns. The proposed models do better than the benchmark whenever the black line is below the red one. We intentionally started these plots before July 2007, allowing the reader to see what happens both before and after our chosen start date.

For the first column of panels, referring to the 6-month horizon, we see that the black and red lines move together. There is no single event or moment in which models suddenly get better (or worse) than the benchmark. As seen in the July 2007 panel of Table 4, performance is relatively similar to the benchmark because of that. For the specification with the three variables, in the last row, we do see moments between 2010 and 2014 and then between 2019 and 2021, for which the benchmark did somewhat better than the full model. This gets reflected in a negative R^2 of -0.13 when we start in July 2007, for instance.

The second column of panels from Figure 3 starts shedding light on the poor performance of `D` as a predictor. All models that include that variable (third, fifth, sixth, and seventh columns) display large forecast errors between 2010 and 2014. We note that these forecast errors are due to predictions made with information available up to 2009 if we concentrate on data plotted in 2010, for example. This coincides in timing with the stronger disconnection between `D` and the two other variables documented in Figure 1. If this disconnection is due to some structural change, which is not modeled here, it is natural that it would generate large forecast errors since the model in (4) would have been estimated under a different regime.

The third column confirms our observation about `D` and brings an interesting perspective regarding the dot-com bubble. First, we notice that large errors take place roughly between late 2011 and 2015. For these plots, a forecast error realized, for example, in late 2011 was made with data up to late 2008, around the stronger disconnection between `D` and two other series. The location of these absolute error peaks gets shifted forward in time because we index these

curves by the time they become realized, not the time at which forecasts are made. In any case, the same peaks that appeared with $h = 12$ now appear with $h = 36$. They are also present for $h = 60$ as long as D is included in the regression.

Second, we notice that models conditioning either on CAPE or LTG display absolute forecast error peaks in the early 2000s. That is due to the dot-com bubble burst. An observer with data available only up to early 2000 and making forecasts conditioning on these two variables, as an example, would have predicted high 36-month returns ahead. After all, the stock market was going up in the early 1990s, hand in hand with both CAPE and LTG- which were at their highest levels in that period. This observer would regret his forecast only when it is then revealed, around 2003. The same peaks appear when we concentrate on the last column of panels, which uses $h = 60$, taking place around 2005. Once again, the peaks are shifted forward in time due to our indexing conventions. When making forecasts five years ahead, the optimistic observer from late 2000 would regret her predictions only in 2005. The estimates $\widehat{\beta}_{LTG}$ were indeed positive for this earlier part of our data (as we investigate in greater detail below). When combined with values for LTG at their historical peak, this fact generated overly optimistic forecasts for market returns.

Figure A.4 in Appendix A supports the above interpretation. It plots the signed forecast error across models and horizons as a function of time. If we take the absolute value of the plotted curves, we recover Figure 3. The dotted line represents a reference at zero. Positive values denote forecast errors that are indicative of excessive optimism. One can see that models conditioning on either D or CAPE were caught off-guard by the dot-com burst because the stock market returns were expected to be much higher than they were.

Figure 3 also helps understand why the specification with the three variables fared so badly, even though it nests all models considered here. In some sense, it combines the worst of both cases described above: the conditioning on CAPE and LTG led to excessively high predictions in the earlier part of the sample, and the conditioning on D implied large errors after 2008 when some deeper modification on the dynamics of D might have happened.

5.4 Additional results: R^2_{OOS} and regression coefficients

Appendix B reports two additional diagnostics. Section B.1 plots the full path of $R^2_{OOS}(t_0)$ as the starting date varies, and Section B.2 reports the recursive coefficient paths behind those forecasts.

In Section B.2, we discuss the temporal behavior of regression coefficients as one varies the

training data. For a well-specified statistical model, coefficients should remain largely stable. A few empirical patterns call our attention.

First, the coefficient $\widehat{\beta}_{LTG}$ only becomes negative when sufficiently recent data is incorporated. This effect is particularly pronounced for longer horizons, of 36 and 60 months. Researchers with data prior to the dot-com bubble would see negative coefficients across all horizons, which runs counter to theories of extrapolation and overpricing. For longer horizons, the sign flips after the Great Financial Crisis. This is true for models using `LTG` alone and combined with `CAPE`.

Second, for most possible sample periods and horizons, the coefficient $\widehat{\beta}_{D}$ is statistically indistinguishable from zero. Yu (2011) is published in a particular period in which the coefficient on `D` is negative and statistically significant across multiple horizons. This is not true for earlier samples nor samples that include sufficiently recent data.

6 Conclusion

This paper tests whether analyst long-term growth forecasts (`LTG`) and disagreement predict aggregate market returns beyond what is captured by the cyclically adjusted price-earnings ratio (`CAPE`). Our main finding is that they do not.

In sample, the significant predictive power of `LTG` reported in Bordalo et al. (2024) does not survive the substitution of `CAPE` for the spot PE ratio they employ. The same holds for analyst disagreement. Rolling-window estimates reveal that the `LTG`–return relationship is unstable across subperiods, with coefficients that change sign and magnitude—particularly around 2008. Out of sample, recursive forecasts across multiple start dates and horizons confirm that `LTG` and disagreement add little beyond the historical mean and never meaningfully outperform `CAPE` alone.

These results do not challenge the evidence that subjective beliefs matter for price formation. The strong contemporaneous correlation between `LTG` and `CAPE`, which we confirm, supports the view that analyst expectations are reflected in current stock prices. Our finding is that this same correlation leaves `LTG` with no incremental information for predicting *future* returns beyond what the valuation ratio already captures. We also document that analyst disagreement exhibits a notable behavioral shift after 2008: its correlation with `LTG` and `CAPE` weakens, and out-of-sample forecast errors grow substantially when models estimated on pre-crisis data are evaluated in later periods.

We highlight two caveats. First, our study uses monthly data starting in the early 1980s, which is a relatively short sample for highly persistent processes. As we demonstrate, researchers working with truncated subsets of this data would likely have reached different conclusions. This underscores the importance of comprehensive samples and motivates complementary cross-sectional analyses, which we defer to future work. Second, we rely solely on I/B/E/S data to maintain a close connection with the previous literature rather than incorporating other disagreement measures as in Huang et al. (2021).

Finally, we underscore that our evidence does *not* point to fundamental flaws in either the classical argument from Miller (1977) or the extrapolation framework from Bordalo et al. (2024). It *does* show, however, that the return predictability evidence previously used to support these theories is more fragile than originally reported.

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Tables and Figures for the Main Text

Table 1: Summary statistics for the whole sample (December 1981 - December 2022). SD denotes the standard deviation of each variable, while we reserve the notation "Px" for the x-th quantile. The last column reports the number of observations.

	Mean	SD	P5	P25	Median	P75	P95	N
R_6	0.06	0.11	-0.12	0.00	0.06	0.13	0.23	493
R_{12}	0.13	0.17	-0.18	0.05	0.14	0.23	0.39	493
R_{36}	0.44	0.35	-0.23	0.30	0.45	0.62	1.08	493
R_{60}	0.84	0.59	-0.07	0.44	0.90	1.17	1.96	493
CAPE	23.64	8.30	9.61	18.09	23.95	28.38	38.41	493
LTG	13.50	2.11	11.37	12.16	12.93	13.87	18.00	493
D	3.61	0.63	2.79	3.10	3.47	4.09	4.68	493

Table 2: Each panel displays estimates for regression coefficients from (4). All variables are standardized before the regressions. The dependent variables are the 12-, 36-, and 60-month market index returns. Different panels cover different sub-samples. We report t-statistics inside the parentheses. Stars denote significance at 10%, 5%, and 1% confidence levels. Standard errors are computed with a HAC-adjusted estimator following Newey and West (1987) with h lags. The last rows of each table report the number of observations in each regression, and the associated adjusted R^2.

Panel A: Full-Sample (1981-2022)

	R_{12}						R_{36}						R_{60}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
LTG	-0.34***		-0.27**	-0.19		-0.17	-0.37		-0.33	-0.01		-0.01	-0.45***		-0.42***	0.02		0.01
	(-2.97)		(-2.30)	(-0.90)		(-0.88)	(-1.56)		(-1.49)	(-0.03)		(-0.03)	(-5.71)		(-4.76)	(0.13)		(0.06)
D		-0.32**	-0.24**		-0.22*	-0.21*		-0.27	-0.20		-0.10	-0.10		-0.28	-0.22		-0.08	-0.08
		(-2.14)	(-2.07)		(-1.67)	(-1.75)		(-1.16)	(-0.91)		(-0.42)	(-0.42)		(-1.18)	(-0.94)		(-0.29)	(-0.27)
CAPE				-0.23	-0.28	-0.16				-0.55***	-0.52**	-0.52***				-0.70***	-0.66***	-0.66**
				(-0.85)	(-1.55)	(-0.66)				(-3.85)	(-2.29)	(-2.87)				(-3.12)	(-4.36)	(-2.34)
N	481	481	481	481	481	481	457	457	457	457	457	457	433	433	433	433	433	433
R^2	0.117	0.103	0.170	0.145	0.168	0.182	0.129	0.068	0.163	0.288	0.298	0.296	0.184	0.067	0.225	0.421	0.426	0.425

Panel B: Same sample as Bordalo et al. (2024) (1981-2015)

	R_{12}						R_{36}						R_{60}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
LTG	-0.32**		-0.26**	-0.10		-0.09	-0.47***		-0.41***	-0.02		-0.01	-0.47***		-0.39***	0.17		0.22
	(-2.40)		(-2.12)	(-0.38)		(-0.40)	(-3.19)		(-3.47)	(-0.09)		(-0.04)	(-4.15)		(-2.89)	(0.51)		(0.60)
D		-0.35*	-0.30*		-0.27	-0.27		-0.36	-0.28		-0.22	-0.22		-0.38	-0.26		-0.19	-0.22
		(-1.70)	(-1.94)		(-1.52)	(-1.61)		(-1.03)	(-1.21)		(-0.74)	(-0.74)		(-1.40)	(-1.19)		(-0.68)	(-0.71)
CAPE				-0.31	-0.31	-0.25				-0.59**	-0.55***	-0.54*				-0.84***	-0.66***	-0.82***
				(-0.87)	(-1.65)	(-0.81)				(-2.10)	(-3.00)	(-1.70)				(-2.84)	(-4.12)	(-2.62)
N	397	397	397	397	397	397	373	373	373	373	373	373	349	349	349	349	349	349
R^2	0.103	0.121	0.187	0.147	0.211	0.213	0.198	0.120	0.265	0.339	0.382	0.380	0.199	0.130	0.248	0.465	0.484	0.501

Panel C: Same sample as Yu (2011) (1981-2007)

	R_{12}						R_{36}						R_{60}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
LTG	-0.41**		-0.07	-0.27		0.38	-0.65***		-0.33	-0.42		0.47**	-0.63***		-0.45	-0.13		0.68***
	(-2.52)		(-0.43)	(-0.56)		(1.18)	(-3.59)		(-1.27)	(-1.09)		(2.13)	(-3.06)		(-1.57)	(-0.20)		(4.34)
D		-0.53***	-0.48**		-0.44***	-0.61***		-0.68***	-0.45		-0.51*	-0.72**		-0.57***	-0.26		-0.35	-0.66***
		(-3.70)	(-2.56)		(-2.72)	(-3.00)		(-3.30)	(-1.38)		(-1.88)	(-2.22)		(-3.70)	(-0.96)		(-1.39)	(-2.80)
CAPE				-0.18	-0.21	-0.44				-0.28	-0.41**	-0.72***				-0.61	-0.58**	-1.03***
				(-0.30)	(-0.99)	(-1.33)				(-0.73)	(-2.14)	(-4.83)				(-0.98)	(-2.54)	(-4.94)
N	277	277	277	277	277	277	253	253	253	253	253	253	229	229	229	229	229	229
R^2	0.170	0.281	0.281	0.178	0.313	0.337	0.378	0.416	0.460	0.396	0.538	0.562	0.326	0.267	0.351	0.419	0.504	0.557

Table 3: The table reports coefficient estimates for the regressions in (5) and (6). We define $\Delta x_t \equiv x_t - x_{t-12}$ for any variable x_t. We display t-statistics in the parentheses. Standard errors are computed using the HAC estimator from Newey and West (1987). The sample used ranges from December 1981 to December 2022. Stars denote significance at 10%, 5%, and 1% confidence levels.

	$CAPE_t$				$\Delta CAPE_t$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$CAPE_{t-12}$	0.85***				-0.32			
	(3.64)				(-0.63)			
LTG_t		0.68***		0.62***				
		(4.88)		(5.32)				
D_t			0.37*	0.18				
			(1.91)	(1.37)				
ΔLTG_t						0.63***		0.64***
						(4.34)		(4.00)
ΔD_t							0.39**	-0.02
							(2.45)	(-0.15)
N	481	493	493	493	481	481	481	481
R^2	0.785	0.455	0.132	0.483	0.097	0.400	0.151	0.399

Table 4: Out-of-sample R^2_{OOS} evaluated across different start dates $t_0 \in \{1990, 1995, 2000, 2007\}$. Each panel reports $R^2_{OOS}(t_0, h)$ for seven predictor specifications and four forecasting horizons. Clark-West significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. "NA" denotes infeasible horizon/start-date combinations (insufficient post-t_0 observations for $h = 60$ with $t_0 = 1990$).

Predictors	h=6	h=12	h=36	h=60
t_0 **= Jan 1990**				
CAPE	0.01**	-0.06	-0.42	NA
LTG	0.01	-0.12	-1.66	NA
D	-0.01	-0.08	-0.35	NA
CAPE + LTG	-0.21	-0.50	-1.87	NA
CAPE + D	-0.07	-0.24	-0.54	NA
LTG + D	-0.03	-0.25	-1.69	NA
CAPE + LTG + D	-0.28	-0.56	-1.83	NA
t_0 **= Jan 1995**				
CAPE	-0.02	-0.09	-0.54	-0.95
LTG	0.02	-0.13	-1.81	-3.30
D	0.01	-0.00	-0.38	-0.18
CAPE + LTG	-0.10	-0.40	-2.13	-4.84
CAPE + D	-0.12	-0.31	-0.62	-0.56
LTG + D	-0.01	-0.12	-1.82	-2.93
CAPE + LTG + D	-0.22	-0.61	-2.08	-3.22
t_0 **= Jan 2000**				
CAPE	0.08**	0.12	-0.15	-0.09
LTG	0.03	-0.15	-2.26	-3.83
D	0.01	-0.01	-0.42	-0.19
CAPE + LTG	0.01	-0.26	-2.13	-4.30
CAPE + D	0.06*	0.09*	-0.25	0.22
LTG + D	-0.00	-0.16	-2.23	-3.41
CAPE + LTG + D	-0.06	-0.37	-2.10	-3.00
t_0 **= Jul 2007**				
CAPE	0.04	0.08	0.12	0.36
LTG	0.01	0.04	-0.51	-0.22
D	-0.09	-0.32	-1.51	-0.92
CAPE + LTG	0.03	0.06	-0.17	0.21
CAPE + D	-0.06	-0.19	-0.77	0.08
LTG + D	-0.09	-0.28	-1.35	-0.38
CAPE + LTG + D	-0.13	-0.48	-2.18	-1.98

Figure 1: Time-series evolution for the consensus growth forecast `LTG`, the associated disagreement `D`, and the cyclically adjusted price-earnings ratio `CAPE`. All variables have been normalized to a common scale for plotting. The sample spans December 1981 to December 2022. The shaded bars track NBER recessions.

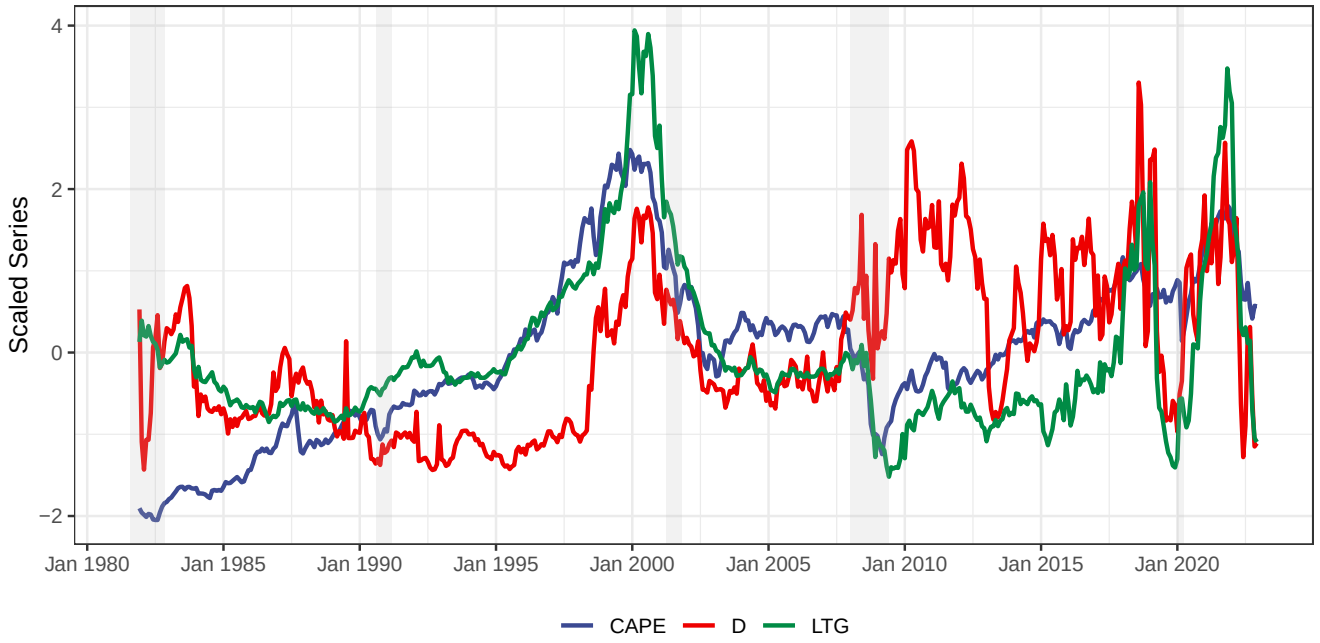


Figure 2: We compute the 120-month rolling correlation between variables. The curves are indexed by the last month used in the computation, so they all start in December 1991. Shaded bars represent NBER recessions.

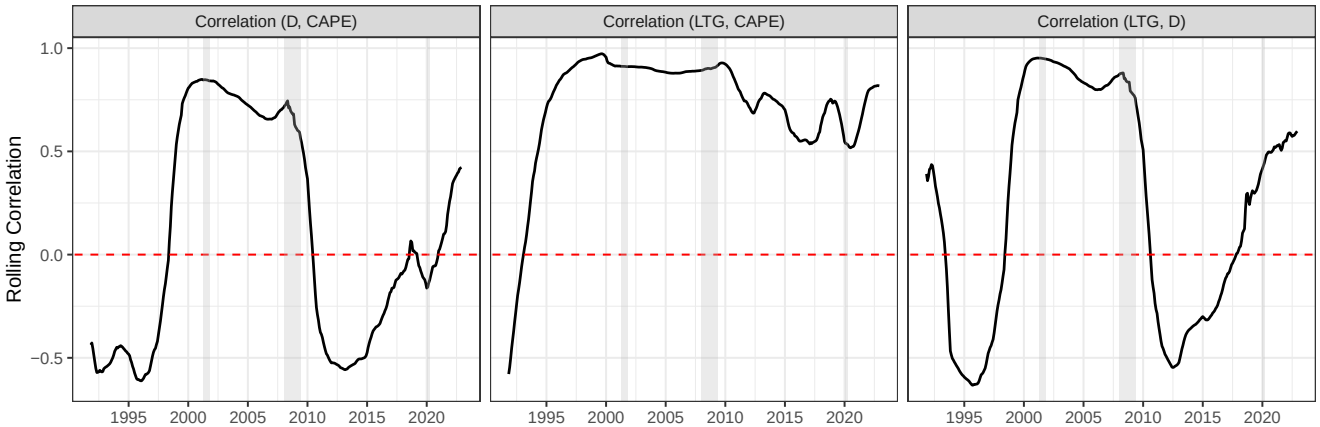
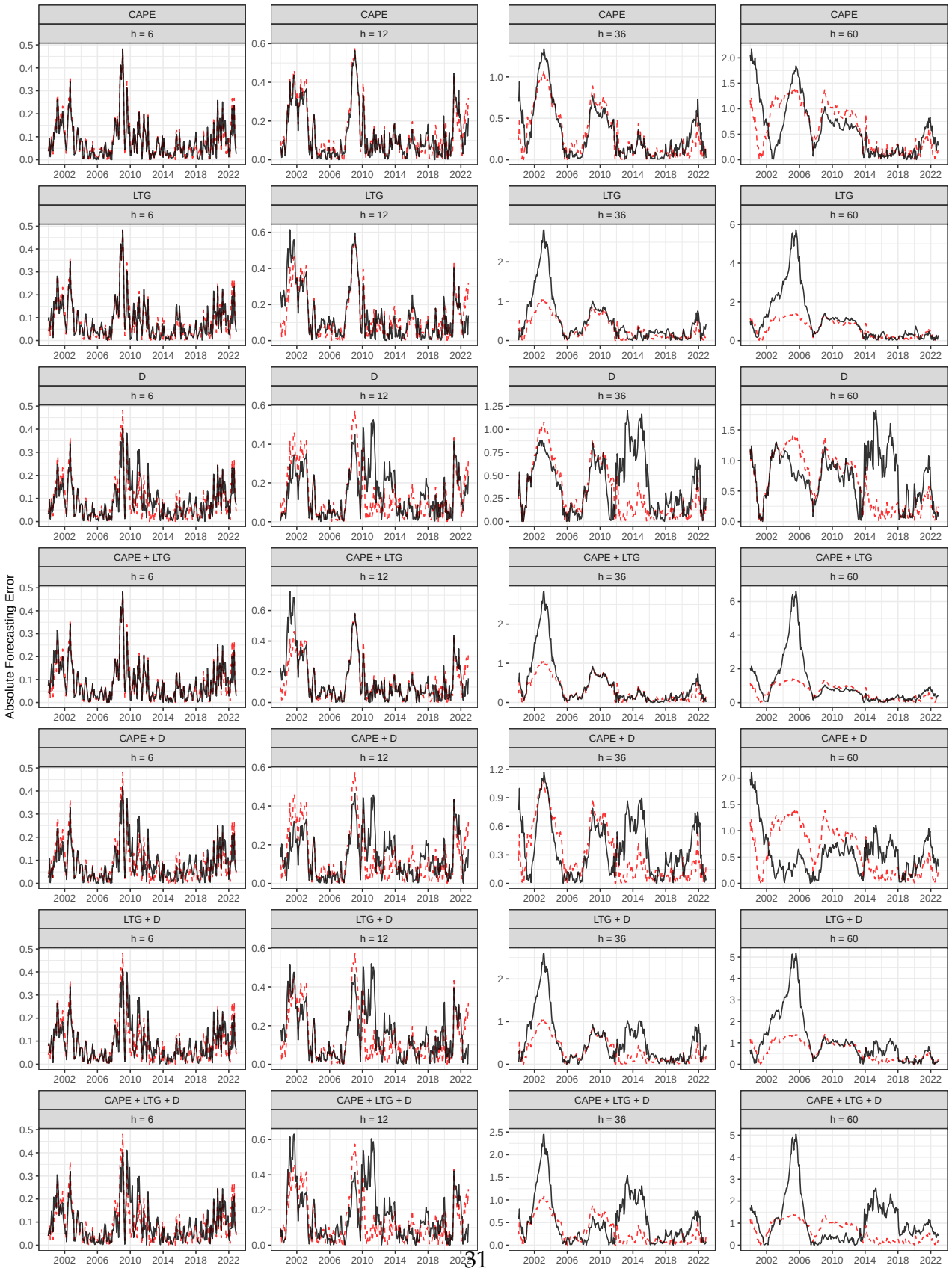


Figure 3: Each plot depicts the evolution of $\left|e_{t,h}^{(M)}\right|$ for different models (rows) and different horizons (columns). The solid black line represents the absolute forecast error. The red dashed line represents the error from the constant-only benchmark model. The index convention is such that the points are plotted as they are realized. See the main text for details.



Supplementary Appendix

A Additional Figures

Figure A.1: Autocorrelation function for the variables from I/B/E/S and for `CAPE`. We plot up to lag 36. We use the full sample for estimation (December 1981 - December 2022).

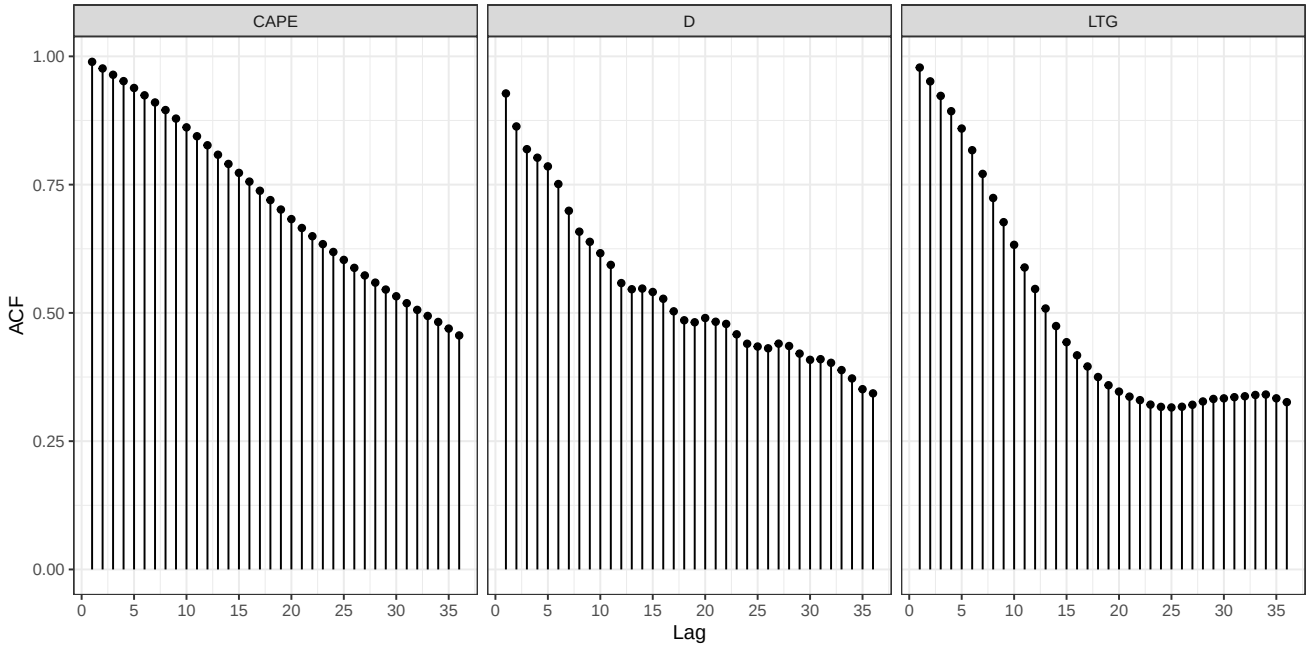


Figure A.2: We plot the evolution of $X_{t+12} - X_t$ for $X \in \{D, LTG, CAPE\}$. Data range from December 1981 to December 2022. Shaded areas represent NBER recessions. We normalized variables before plotting so they could be shown on the same scale.

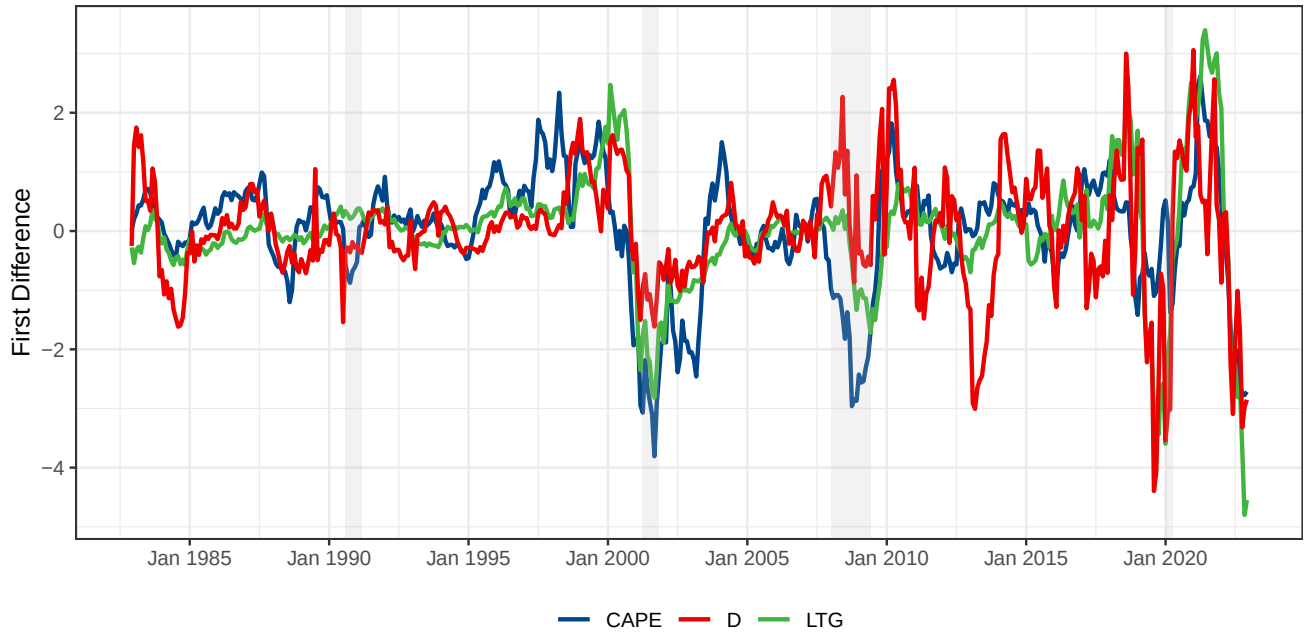


Table A.1: Correlation matrix: belief measures and sentiment indices. Upper triangle reports Pearson correlations; lower triangle reports Spearman rank correlations. Sample period: Dec 1981 – Dec 2020 ($N = 469$ months).

	LTG	D	CAPE	BW Sentiment	HLW Index
LTG	1.00	0.12	0.29	0.41	0.36
D	-0.00	1.00	0.32	-0.22	0.32
CAPE	0.22	0.31	1.00	-0.04	0.18
BW Sentiment	0.49	-0.36	-0.06	1.00	0.10
HLW Index	0.30	0.29	0.10	-0.01	1.00

Notes:

Upper triangle reports Pearson correlations; lower triangle reports Spearman rank correlations. Sample period: Dec 1981 – Dec 2020 (N = 469 months). BW Sentiment is the Baker and Wurgler (2006) index orthogonalized to macroeconomic conditions. HLW Index is the PLS investor sentiment index from Huang et al. (2021).

Figure A.3: We plot all the time series used in the analysis in their original scales. The sample spans December 1981 to December 2022. Shaded areas represent NBER recessions.

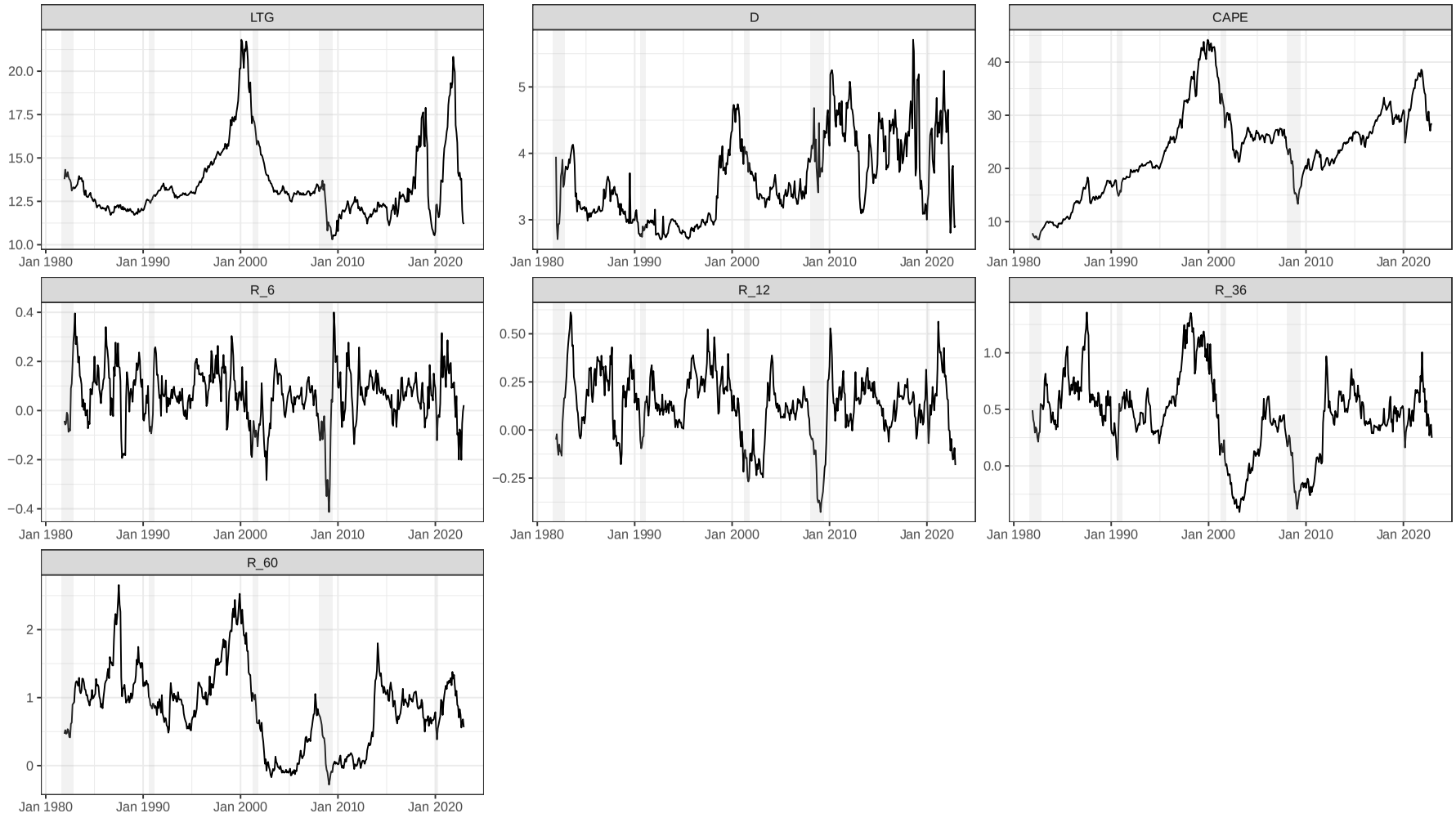
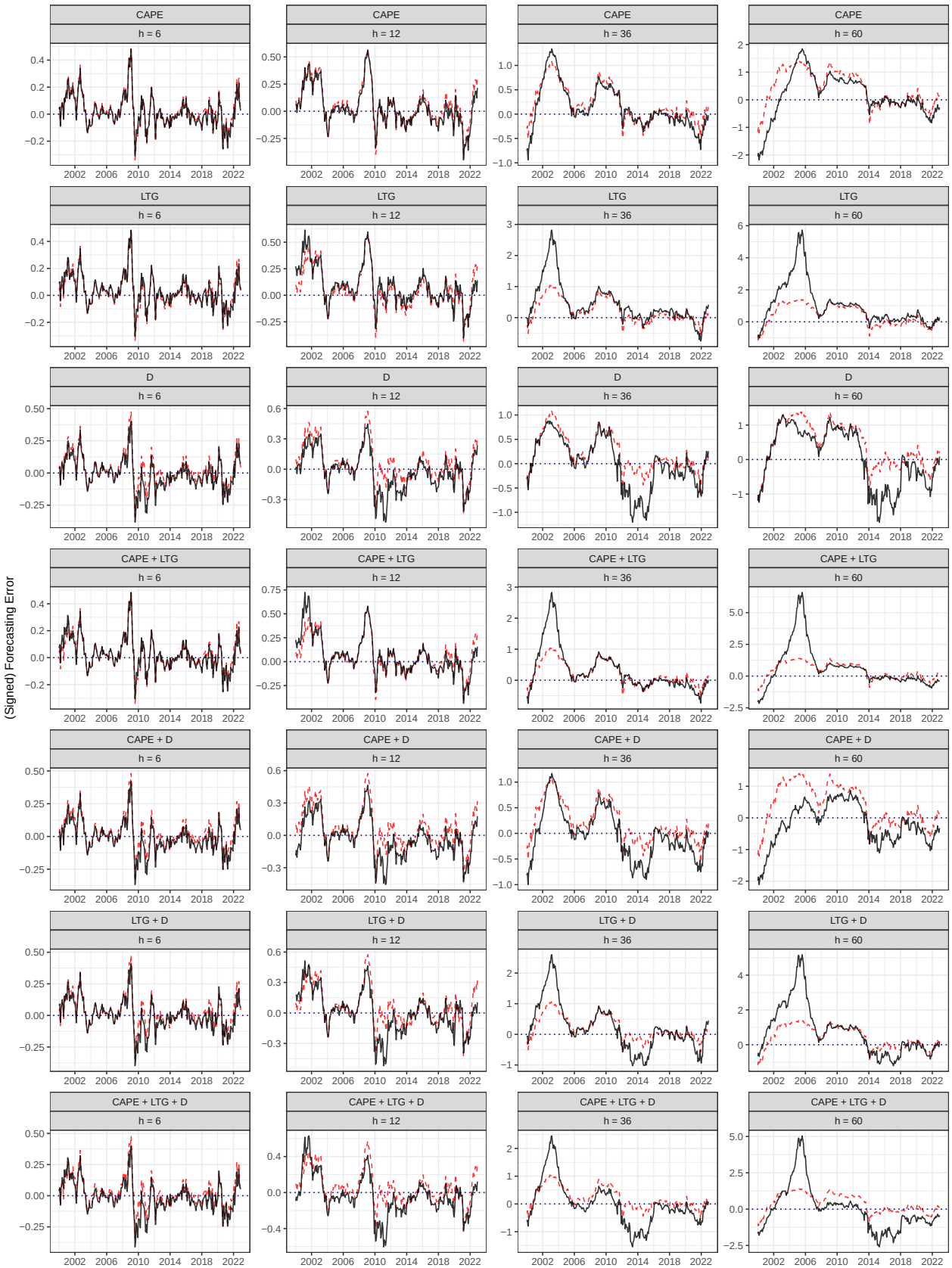


Figure A.4: We show the evolution of $e_{t,h}^{(M)}$. See Figure 3 for details. The only difference is that Figure 3 reports the absolute value of the measures shown here.



B Additional out-of-sample results

B.1 Evolution of R^2_{OOS}

Another way of assessing the sensitivity of different models to the out-of-sample period is through the evolution of $R^2_{OOS}(t_0)$. Figure B.1 follows the same scheme of plotting from Figure 3: each column is devoted to a forecasting horizon h while different models are displayed in different rows. For each point in time t, we plot $R^2_{OOS}(t)$, which is the value we would obtain if we started the out-of-sample period at t. The point corresponding to July 2007, therefore, matches the July 2007 panel of Table 4. In that sense, the figure allows the reader to answer directly the counterfactual question of what the out-of-sample fit would have been had the evaluation window started earlier or later.[16]

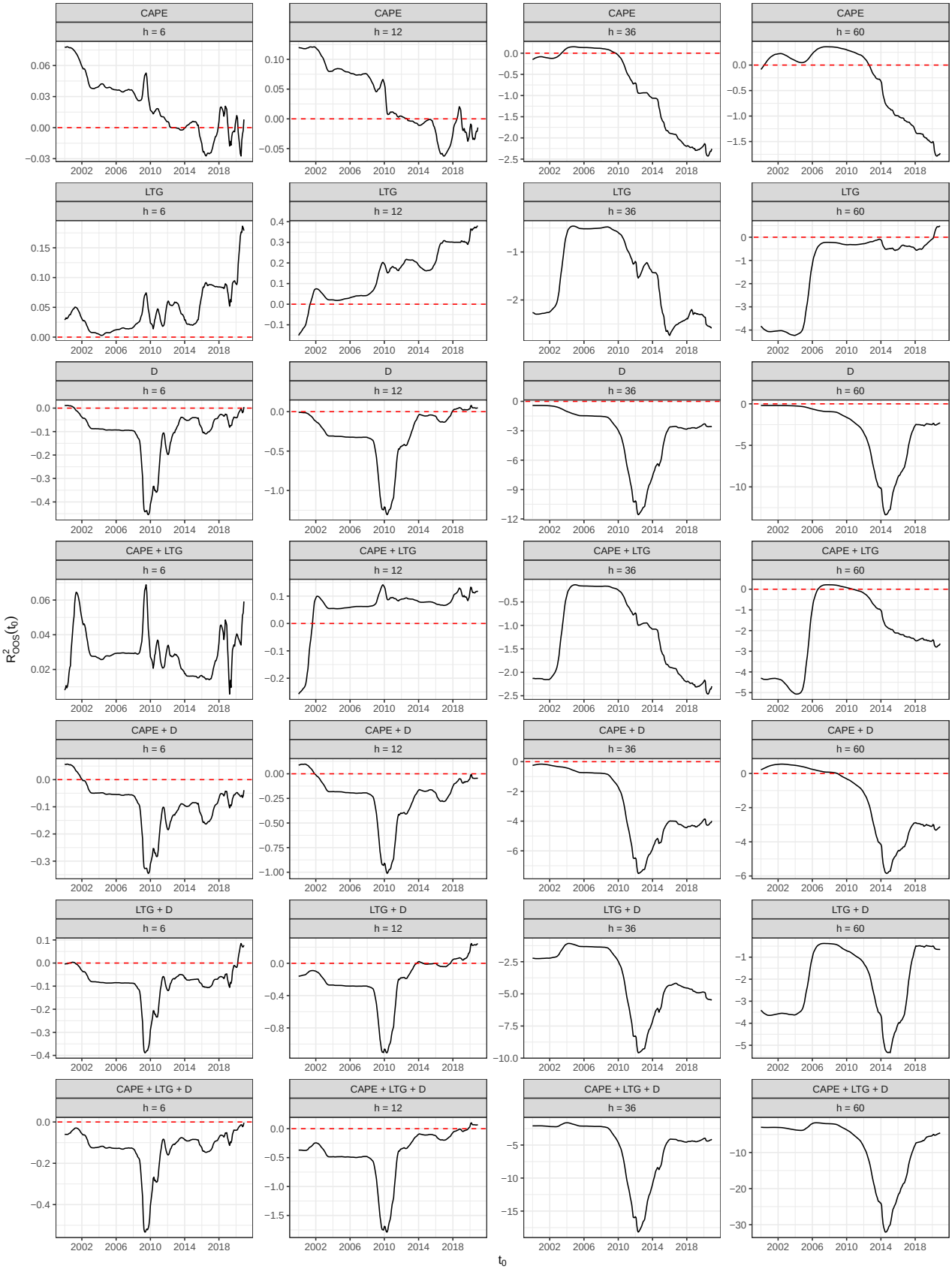
A reader who disagrees with our choice of baseline out-of-sample period can pick her favorite starting point and look up R^2 values in Figure B.1. However, caution is needed as we approach the end of these curves. These last points are computed using far fewer observations, making the out-of-sample R^2 a noisier performance index as we move forward in the horizontal axis. We still believe this picture is useful as a call to caution when picking out-of-sample periods.

There are a few interesting lessons from Figure B.1, providing support for our earlier interpretations. First, we note that all plots displaying models containing `D` have a type of "valley" of predictability, i.e., a sharp drop in out-of-sample R^2. As we have emphasized, the timing coincides with the moment in which `D` started displaying a more jagged path over time. As we move forward and start the sample later, effectively excluding the earlier regime from the out-of-sample period, we see that the out-of-sample R^2 improves. Nonetheless, we rarely find starting dates for which models including `D` display positive short-horizon values, and these gains are neither large nor persistent across horizons. The third row makes this pattern clear.

The inspection of the shorter horizons brings the second pattern to notice. The first row shows that the predictive power of `CAPE` alone decreases if we choose starting points later in the sample. Figure A.4 reveals that this happens because this model overestimated stock market performance slightly less than the sample average during the early 2000s. As we move forward the starting point of our out-of-sample period, these mistakes are no longer relevant for the computation of the performance metric, which then drifts down.

[16] Clark–West p-values for selected start dates are reported in Table 4; in this figure, we omit them in the interest of visual clarity, as most paths are frequently negative.

Figure B.1: We plot the evolution of $R^2_{OOS}(t_0, h)$ over time for different models (rows) and different forecasting horizons (columns). Positive values indicate that a model would have beaten the benchmark if we had started the out-of-sample period at t_0. Negative values imply otherwise.



The exact reverse happens with LTG. Using later starting points, we would find that LTG is a stronger solo predictor. This is because this model had some out-of-sample overestimates in the early 2000s that exceeded the mistakes made by the historical average. The performance metric increases as we discard these earlier observations from the out-of-sample period, moving the starting point further in time. When we combine both measures, we see no apparent drift in the performance metric, aside from an initial sharp increase for $h = 12$. Even there, however, the bivariate model does not deliver a robust improvement over CAPE alone.

The last insight comes from $h = 36$ and $h = 60$. Results are dismal across columns and rows, except possibly for the model that uses only CAPE (first row). For all other models, once more, even if we tried our best to p-hack results, we would almost always find negative values for R^2_{OOS}. If we start the out-of-sample period in 2014, however, it holds without qualification that these profiles of out-of-sample R^2 are uniformly negative.

B.2 Regression Coefficients

Now, the analysis concentrates on the coefficient estimates from (4). Moving forward with our recursive forecasting scheme, we collect the time series of coefficient estimates and associated standard errors.[17] Ideally, we would expect estimated coefficients to remain stable over time if the model is correctly specified. Additionally, we would also expect confidence intervals to shrink as we estimate the same regression with an increasing number of data points. In the interest of brevity, we focus only on the estimates of $\widehat{\beta}_{LTG}$ and $\widehat{\beta}_D$. We analyze how these estimates vary over time for each forecasting horizon and specification.

Figure B.2 reports our results in three panels. The columns in each of these panels display results for different forecasting horizons. The first row is always dedicated to $\widehat{\beta}_{LTG}$, while the second one shows results for $\widehat{\beta}_D$. The first panel focuses on specifications where LTG and D were the only conditioning variables. The second panel reports coefficient estimates when both variables are included in the regression model. The last panel shows estimates when CAPE is included as well. The solid line is the time series of estimated coefficients in all subplots. We index these estimates according to the target date to keep our convention consistent. The curves plotted at date t represent the coefficients used to make a forecast for the random variable $R_{t|t-h}$ using information up to $t - h$. The darker shade represents a 1-standard-error symmetric band around the point estimate, while the lighter shade represents a 2-standard-deviation band. The final value for each solid line corresponds to the full-sample estimates we presented in Table 2.

[17] We use HAC-adjusted standard errors following Newey and West (1987) with h lags.

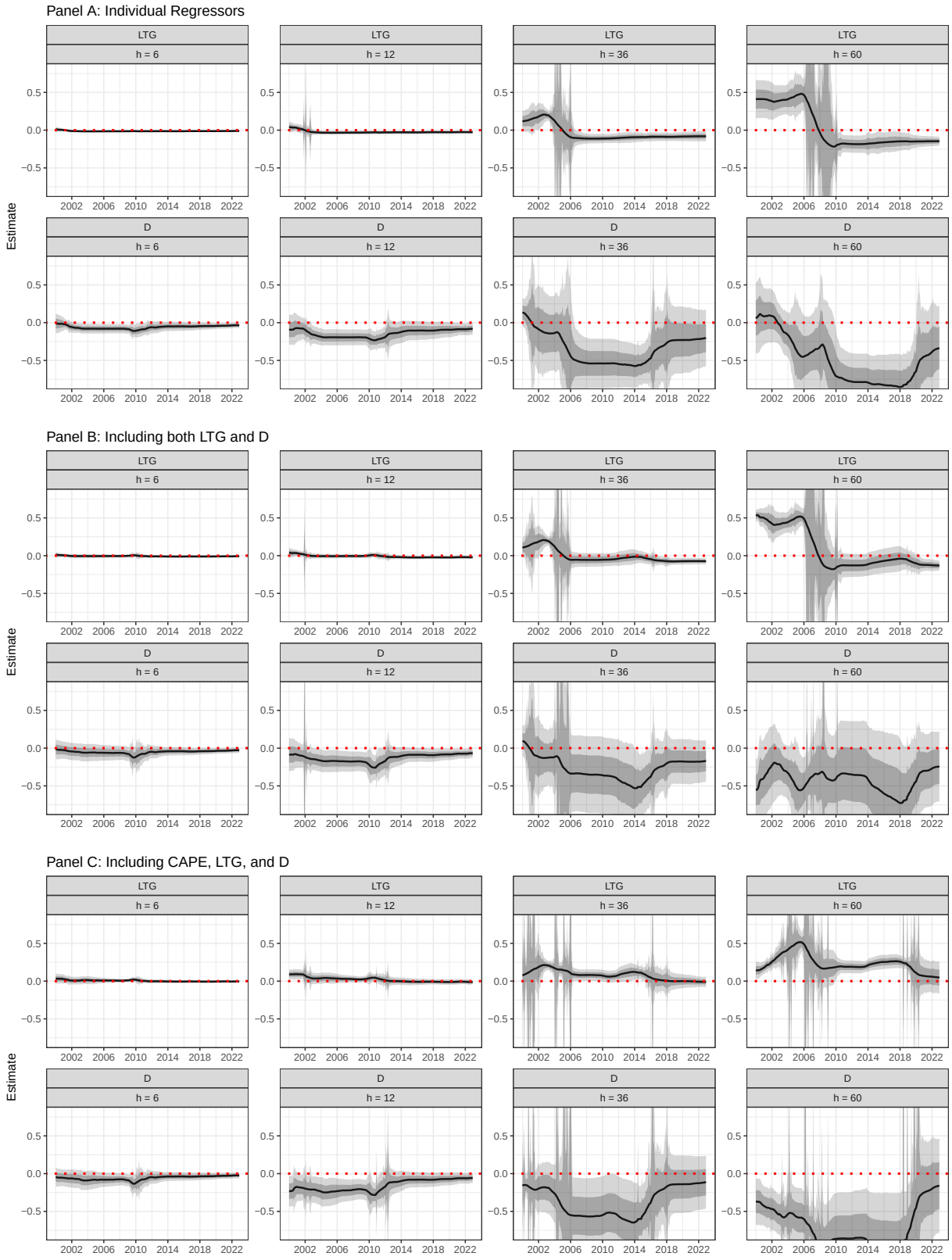
We see two main patterns in this figure. The first one is related to the behavior of $\widehat{\beta}_{LTG}$. When included alone or alongside `D`, the estimates start as positive and migrate to the negative plane with the arrival of more recent data. This pattern gets stronger as we forecast at longer horizons. If Bordalo et al. (2024) were to run the same regression they studied in Table 2 of their paper with data only up to 2003, for example, they would find a different sign for their estimates. We can't rule out that this is just estimation uncertainty getting reduced as more data comes in, but we believe it is important to document this change in sign since it has different implications for models of subjective belief formation.

These positive estimates are closely connected to overly optimistic forecasts generated by these models on the verge of the dot-com bubble burst. Positive estimates for the coefficient on `LTG`, coupled with an ever-rising consensus growth forecast, led to high estimates of market returns. The estimates for β_{LTG} start being negative only after the forecast errors due to the dot-com bubble are realized. Alternatively, when `CAPE` is also included, the estimates are either insignificant or have a positive sign all the time.

The second pattern comes from the evolution of $\widehat{\beta}_{D}$. We note that the point estimates are typically negative, as Yu (2011) found and as we confirmed using his sample. However, the associated light shades often include zero, suggesting imprecise estimation, in particular, as we accumulate more data. In light of the jagged behavior of `D` after 2008, one possible explanation is that the larger samples used to generate the final point estimates mix data from different regimes, making it impossible to estimate the desired parameter due to misspecification. The second explanation is that the coefficient is zero, but due to having a smaller and finite sample, Yu (2011) found negative and significant results. We cannot rule out any of these explanations.

Nevertheless, we notice some "pockets of significance", i.e., data strips that would lead to significant results if a researcher were given only a truncated version of our sample. That can be seen, for example, on the second row of the first panel. In that case, `D` is included alone in these regressions. The estimates start as being positive, although insignificant. As more data comes in, they become negative. During that period, Yu (2011) was published. More data comes in, and significance disappears once again. A similar qualitative phenomenon happens when we control for `LTG` and `CAPE`.

Figure B.2: Each panel shows the evolution of OLS estimates from regression (4). Panel A is dedicated to univariate models. Panel B studies the bivariate model that includes both `LTG` and `D`. Panel C shows the estimates one would get from including `CAPE` alongside the other two measures from I/B/E/S . Darker and lighter shades indicate symmetric 1 and 2 standard-deviation bands around point estimates, computed using the HAC estimator from Newey and West (1987) with h lags.



C Comparing Datasets

We highlight some differences between our empirical approach and the one taken by Bordalo et al. (2024). None of the differences is relevant to our results. We concentrate on two fronts: slightly different definitions of market returns and different choices for the valuation ratio (our `CAPE` versus their spot PE).

C.1 Market Returns

Our empirical analysis used arithmetic market returns (as defined in Equation 1), instead of discounted log-returns as Bordalo et al. (2024). They target the following variable for different horizons h:

$$(\text{Discounted Log-Returns})_t \equiv \sum_{i=1}^{h} \alpha^{(i-1)} r_{t+i\cdot 12} \tag{9}$$

where $r_t \equiv \log(S_t) - \log(S_{t-12})$ is the index annual log-return and they set $\alpha = 0.9779$.

This difference should not be important for our conclusions for two distinct reasons. The first one is that low powers of alpha are fairly close to 1. The second reason is that the dynamics of log-returns are similar to the arithmetic returns for the horizons considered (h = 1,3,5 years).

Figure C.1 plots the arithmetic returns alongside the pure log-returns ($\alpha = 1$) and the discounted log-returns as defined above. We note that the three series are essentially the same for $h = 12$ months and $h = 36$ months. For the longest horizon of 60 months, the arithmetic returns drift away from the two other series only before the early 2000s. The discounting is immaterial regardless of the horizon we consider.

Table C.1 repeats the in-sample forecasting exercise from Table 2 for our full sample (1982-2022) but now using the discounted log-returns instead of arithmetic returns. Our conclusions remain unchanged. As soon as we control for `CAPE`, the significance of the coefficient on `LTG` is gone. Therefore, the in-sample results from Section 4 are not driven by our choice of using arithmetic returns.

Table C.1: Full sample (1982-2022) regressions with **discounted log-returns** instead of arithmetic returns. Standard errors are computed using Newey and West (1987). See Table 2 for details.

	R_{12}						R_{36}						R_{60}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
LTG	-0.36***		-0.29**	-0.23		-0.21	-0.46*		-0.43*	-0.12		-0.12	-0.54***		-0.52***	-0.08		-0.08
	(-3.08)		(-2.38)	(-1.30)		(-1.25)	(-1.85)		(-1.78)	(-0.49)		(-0.48)	(-7.48)		(-6.25)	(-0.32)		(-0.31)
D		-0.32*	-0.24*		-0.22	-0.21*		-0.26	-0.17		-0.08	-0.08		-0.25	-0.18		-0.03	-0.04
		(-1.94)	(-1.92)		(-1.58)	(-1.67)		(-0.98)	(-0.88)		(-0.35)	(-0.34)		(-0.98)	(-0.90)		(-0.12)	(-0.13)
CAPE				-0.19	-0.26	-0.12				-0.52***	-0.57*	-0.49**				-0.68**	-0.73***	-0.67**
				(-0.86)	(-1.52)	(-0.58)				(-3.03)	(-1.88)	(-2.37)				(-2.48)	(-4.71)	(-2.06)
N	480	480	480	480	480	480	456	456	456	456	456	456	432	432	432	432	432	432
R^2	0.126	0.101	0.176	0.144	0.160	0.182	0.201	0.062	0.224	0.340	0.337	0.343	0.271	0.054	0.298	0.500	0.498	0.500

Table C.2: Full sample (1982-2022) results with arithmetic returns and **using log(CAPE) as a control**. In the text, we use CAPE in levels. Standard errors are computed using Newey and West (1987). See table 2 for details.

	R_{12}						R_{36}						R_{60}					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)
LTG	-0.34***		-0.27**	-0.21		-0.18	-0.37		-0.33	-0.15		-0.14	-0.45***		-0.42***	-0.16		-0.16
	(-2.97)		(-2.30)	(-1.22)		(-1.08)	(-1.56)		(-1.49)	(-0.64)		(-0.62)	(-5.71)		(-4.76)	(-1.19)		(-1.07)
D		-0.32**	-0.24**		-0.23	-0.20		-0.27	-0.20		-0.14	-0.12		-0.28	-0.22		-0.11	-0.11
		(-2.14)	(-2.07)		(-1.65)	(-1.63)		(-1.16)	(-0.91)		(-0.54)	(-0.47)		(-1.18)	(-0.94)		(-0.34)	(-0.34)
log(CAPE)				-0.26	-0.29	-0.21				-0.44***	-0.48*	-0.41**				-0.56***	-0.62***	-0.53**
				(-1.03)	(-1.60)	(-0.89)				(-2.90)	(-1.83)	(-2.12)				(-2.93)	(-2.69)	(-2.06)
N	481	481	481	481	481	481	457	457	457	457	457	457	433	433	433	433	433	433
R^2	0.117	0.103	0.170	0.164	0.178	0.199	0.129	0.068	0.163	0.264	0.263	0.275	0.184	0.067	0.225	0.397	0.391	0.407

C.2 Other Variables

Now, we analyze how other variables used in our analyses compare to the ones in Bordalo et al. (2024). The first panel in Figure C.2 displays the LTG series we used and the series used by Bordalo et al. (2024) in their empirical analysis. The two series are essentially indistinguishable.

The second panel compares the price-earnings series used in each analysis. We use the cyclically adjusted price-earnings ratio CAPE from Robert Shiller's website, while they use the (log of the) pure price-earnings ratio. The pure series (blue) is more volatile than the adjusted series (red) and shows a large spike during the 2008 financial crisis. The CAPE series is constructed as the ratio of inflation-adjusted prices to a 10-year average of inflation-adjusted earnings (see Campbell and Shiller (1988)), while the pure series is the ratio of the index price to average earnings at a given point in time.

We select the CAPE series as our price-earnings measure because it is the standard choice in the market-return forecasting literature, with evidence of its predictive power for the stock market at long horizons in both the US and abroad (Keimling, 2016; Jivraj and Shiller, 2017). This choice aligns with our primary objective of assessing the incremental forecasting capability of LTG and D variables.

Additionally, we use the CAPE ratio in levels, not in log points. This choice is, once again, immaterial. Table C.2 repeats our main in-sample exercise from Table 2 but now using $\log(\texttt{CAPE})$ as a control. We use arithmetic returns, as in the main text. We see similar quantitative results. As soon as we control for the log of the adjusted price-earnings ratio, we cannot reject that the coefficient on both LTG and D is insignificant.

Figure C.1: For each different horizon, we plot arithmetic returns (as defined in equation 1), log returns, and discounted log returns (as in (9)). In all cases, we considered the value-weighted CRSP index as our notion of overall market portfolio.

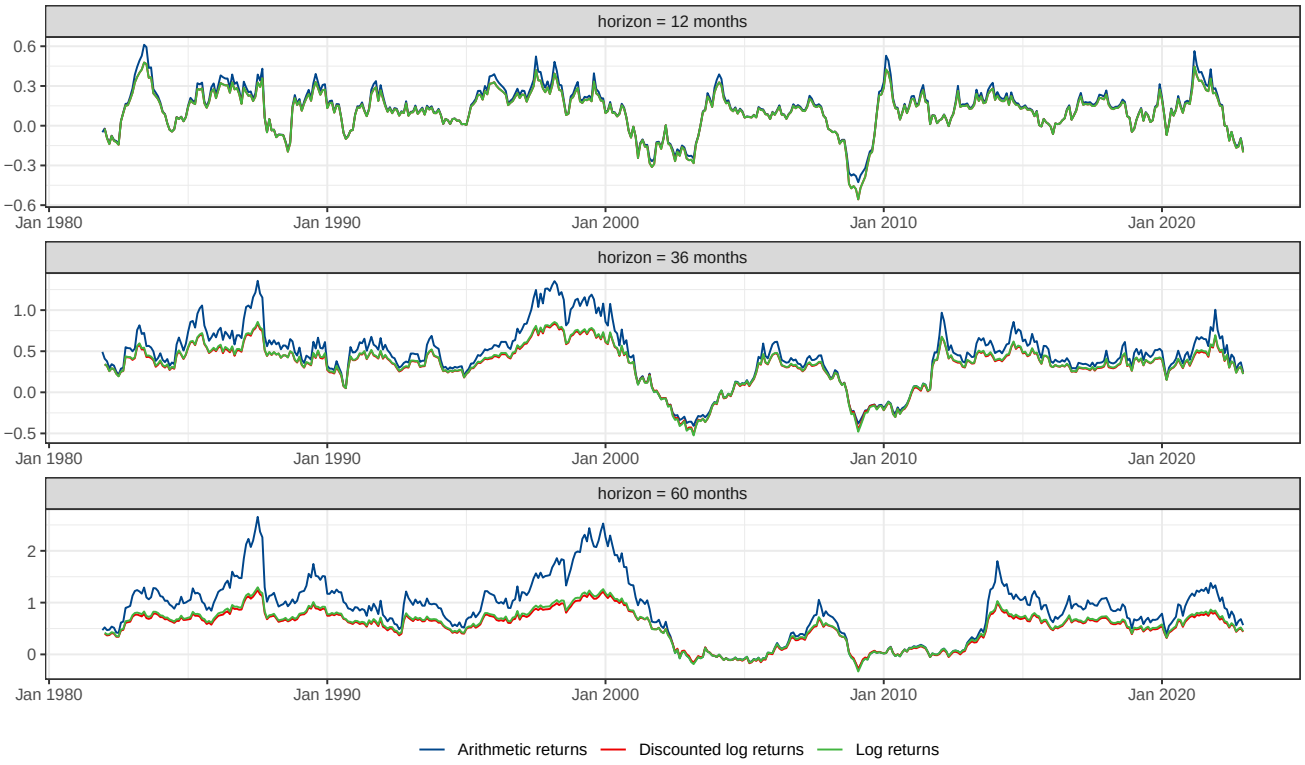


Figure C.2: The first panel displays the `LTG` series constructed from I/B/E/S data. The second panel contrasts the spot PE ratio (blue) and the `CAPE` ratio (red). The third panel displays the logarithm of these series.

